

THE LAX–PHILLIPS–PAVLOV–FADDEEV ASYMPTOTIC CRITERION FOR THE RIEMANN HYPOTHESIS A CONVENTION-FIXED PROBLEM SPECIFICATION AND SOLVER HANDOFF

OPENAI CODEX

ABSTRACT. This document fixes a mathematically unambiguous target corresponding to the Pavlov–Faddeev asymptotic criterion quoted in later accounts of the Lax–Phillips scattering theory of the automorphic wave equation. It defines the reduced positive energy space, the interaction semigroup, its generator and resolvent, and the precise regularized operator norm. In the convention adopted here, the quoted RH-equivalence is

$$\text{RH} \iff \limsup_{t \rightarrow \infty} \frac{1}{t} \log \|Z(t)(B + iI)^{-1}\|_{E \rightarrow E} = -\frac{1}{4}.$$

The resolvent factor is essential. Equivalently, the estimate measures the operator norm of $Z(t)$ from the generator domain with its graph norm into the ordinary energy space. The unregularized norm $\|Z(t)\|_{E \rightarrow E}$ and the fixed-vector quantity $\|Z(t)f\|_E$ are different objects and are recorded separately. Conditional on the physical-to-model eigenvalue correspondence, the implication from the regularized upper bound to RH is proved below. The reverse implication is attributed to Pavlov–Faddeev by a 2017 survey but is not reconstructed from the 1972/1975 primary paper here. Accordingly, this handoff separates three tasks: verifying the historical theorem, reproving its conditional analytic direction, and deciding the regularized estimate unconditionally. Only the last task would decide RH.

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1. THE TARGET STATEMENT IN ONE BOX

The notation in this box is defined in full in Sections 3–6.

Let $(\mathcal{K}, \langle \cdot, \cdot \rangle_E)$ be the reduced Lax–Phillips interaction Hilbert space for the modular automorphic wave equation. Let $Z = (Z(t))_{t \geq 0}$ be its interaction semigroup and let B be the ordinary C_0 -semigroup generator, so that $Z(t) = e^{tB}$. Assume the explicitly listed energy-space, scattering, resolvent, and zeta-eigenvalue identification hypotheses in this document. In particular, assume $-i \in \rho(B)$ and put

$$M(t) := \left\| Z(t)(B + iI)^{-1} \right\|_{E \rightarrow E}, \quad \omega_{\text{reg}}(B) := \limsup_{t \rightarrow \infty} \frac{1}{t} \log M(t).$$

The mathematical target is $\omega_{\text{reg}}(B) = -1/4$. The historical equivalence quoted by Takhtajan et al. is

$$\text{RH} \iff \omega_{\text{reg}}(B) = -\frac{1}{4}.$$

Under the zeta-eigenvalue identification, $\omega_{\text{reg}}(B) \geq -1/4$ unconditionally, so the right-hand condition can equivalently be written

$$\text{RH} \iff \omega_{\text{reg}}(B) \leq -\frac{1}{4}.$$

The norm is

$$M(t) = \sup_{0 \neq h \in \mathcal{K}} \frac{\|Z(t)(B + iI)^{-1}h\|_E}{\|h\|_E} = \sup_{0 \neq f \in \text{Dom } B} \frac{\|Z(t)f\|_E}{\|(B + iI)f\|_E}.$$

The last equality is the most important norm specification in this document: the source norm is a graph norm on $\text{Dom } B$, while the target norm is the ordinary positive energy norm.

2. ARITHMETIC NORMALIZATION

Definition 2.1 (Zeta and completed zeta). For $\text{Re } s > 1$ let

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s}.$$

Its meromorphic continuation has a simple pole at $s = 1$. Define

$$\xi(s) := \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s).$$

Then ξ is entire and satisfies

$$\xi(s) = \xi(1-s) = \overline{\xi(\bar{s})}.$$

A *nontrivial zero* is a zero $\rho = \beta + i\gamma$ of ξ .

Definition 2.2 (Riemann hypothesis). The Riemann hypothesis is the assertion

$$\text{RH} : \quad \xi(\rho) = 0 \implies \text{Re } \rho = \frac{1}{2}.$$

The multiset of zeros is invariant under $\rho \mapsto \bar{\rho}$ and $\rho \mapsto 1 - \rho$.

The factor $1/4$ in the target asymptotic is forced by the following model normalization, once the physical-to-model identification in Hypothesis 8.2 is available:

$$a = \frac{\rho}{2} \quad \mapsto \quad \lambda_{\rho} = -\bar{a} = -\frac{\bar{\rho}}{2}, \quad \text{Re } \lambda_{\rho} = -\frac{\beta}{2}.$$

Thus the critical line $\beta = 1/2$ maps to the semigroup spectral line $\operatorname{Re} \lambda = -1/4$.

3. THE AUTOMORPHIC WAVE ENERGY SPACE

3.1. The spatial operator and wave equation. Let

$$X = \operatorname{PSL}_2(\mathbb{Z}) \backslash \mathbb{H}, \quad d\mu(z) = \frac{dx dy}{y^2}, \quad z = x + iy,$$

and let

$$\Delta_X = -y^2(\partial_x^2 + \partial_y^2)$$

denote the nonnegative self-adjoint Friedrichs realization of the automorphic Laplacian in $L^2(X, d\mu)$. Following the Lax–Phillips normalization, set

$$A := \Delta_X - \frac{1}{4}I.$$

The corresponding automorphic wave equation is

$$(1) \quad u_{tt} + Au = 0.$$

The absolutely continuous spectrum of A starts at 0. Below-threshold modes and threshold null directions are responsible for indefiniteness or degeneracy of the raw energy form. Positive-energy cusp forms are removed for a different reason: they are point-spectrum standing waves rather than scattering states.

3.2. Energy form and positive reduction. Let P_{ac} be the absolutely continuous spectral projection of A , put

$$\mathfrak{h}_c := \operatorname{Ran} P_{ac}, \quad A_c := A|_{\mathfrak{h}_c},$$

and note that $A_c \geq 0$. For sufficiently regular Cauchy data $f = (f_0, f_1)$ and $g = (g_0, g_1)$ in this sector, define

$$(2) \quad E(f, g) := \left\langle A_c^{1/2} f_0, A_c^{1/2} g_0 \right\rangle_{L^2(X)} + \langle f_1, g_1 \rangle_{L^2(X)},$$

Equivalently, on a common operator core,

$$E(f, g) = \langle A_c f_0, g_0 \rangle_{L^2(X)} + \langle f_1, g_1 \rangle_{L^2(X)}.$$

Define

$$\mathfrak{h}_{A_c}^1 := \overline{\operatorname{Dom}(A_c^{1/2}) / \ker A_c} \left\| A_c^{1/2} \cdot \right\|_2$$

and set

$$(3) \quad \mathcal{H}_E^c := \mathfrak{h}_{A_c}^1 \oplus \mathfrak{h}_c, \quad \|(f_0, f_1)\|_E^2 := \left\| A_c^{1/2} f_0 \right\|_2^2 + \|f_1\|_2^2.$$

Thus

$$\|f\|_E := E(f, f)^{1/2}, \quad f \in \mathcal{H}_E^c.$$

This spectral construction explicitly removes the point spectrum. It is a modern convention-fixed realization of the no-standing-waves energy space. The historical construction starts from an auxiliary positive completion of an indefinite energy form and then passes through primed and double-primed quotients/subspaces. The identification of those historical objects with (3) is an input, not a consequence of notation.

Hypothesis 3.1 (Energy-space identification). There is an energy-unitary identification between (3) and the final positive continuous Pavlov–Faddeev/Lax–Phillips energy space on which their interaction semigroup and regularized estimate are defined.

Remark 3.2 (Norm convention). Throughout this document $\|f\|_E$ is the square root of the conserved quadratic energy, as in Lax's Hilbert-space convention. Every operator norm is induced by this Hilbert norm.

3.3. The wave group. Functional calculus gives a unitary group $U(t)$ on $\mathcal{H}_E^c$:

$$U(t) \begin{pmatrix} f_0 \\ f_1 \end{pmatrix} = \begin{pmatrix} \cos(tA_c^{1/2})f_0 + A_c^{-1/2} \sin(tA_c^{1/2})f_1 \\ -A_c^{1/2} \sin(tA_c^{1/2})f_0 + \cos(tA_c^{1/2})f_1 \end{pmatrix}.$$

The formula is interpreted by spectral calculus at the threshold. It satisfies

$$U(t+s) = U(t)U(s), \quad U(t)^* = U(-t), \quad \|U(t)f\|_E = \|f\|_E.$$

4. INCOMING, OUTGOING, AND INTERACTION SPACES

Let $\mathcal{D}_-, \mathcal{D}_+ \subset \mathcal{H}_E^c$ denote the final double-primed (causal, modified) incoming and outgoing subspaces of the automorphic Lax–Phillips construction. Rather than hide their special properties in an unstated convention, we isolate exactly what is used.

Hypothesis 4.1 (Orthogonal automorphic scattering system). The subspaces $\mathcal{D}_\pm$ are closed and satisfy

$$\begin{aligned} U(t)\mathcal{D}_+ &\subset \mathcal{D}_+ & (t \geq 0), & & U(t)\mathcal{D}_- &\subset \mathcal{D}_- & (t \leq 0), \\ \bigcap_{t \in \mathbb{R}} U(t)\mathcal{D}_\pm &= \{0\}, & \overline{\bigcup_{t \in \mathbb{R}} U(t)\mathcal{D}_\pm} &= \mathcal{H}_E^c, \\ \mathcal{D}_- &\perp_E \mathcal{D}_+. \end{aligned}$$

Moreover, the operators defined in (4) below map $\mathcal{K}$ into itself, form a C_0 contraction semigroup, and obey the compression identity following that display.

The adjective modified records the causal correction and removal of the exceptional/standing-wave directions. These orthogonality and compression properties are special to the final automorphic construction; they are not asserted for arbitrary incoming/outgoing subspaces.

Let $P_{\mathcal{D}_\pm}^E$ be the energy-orthogonal projections and set

$$Q_\pm := I - P_{\mathcal{D}_\pm}^E, \quad \mathcal{K} := \mathcal{H}_E^c \ominus_E (\mathcal{D}_- \oplus \mathcal{D}_+).$$

The Lax–Phillips interaction semigroup is

$$(4) \quad Z(t) := Q_+ U(t) Q_-|_{\mathcal{K}}, \quad t \geq 0.$$

Under Hypothesis 4.1,

$$Z(t) = P_{\mathcal{K}}^E U(t)|_{\mathcal{K}}.$$

The family $Z = (Z(t))_{t \geq 0}$ is then a strongly continuous contraction semigroup on $(\mathcal{K}, \|\cdot\|_E)$.

Some sources denote the final reduced objects by $\mathcal{H}'_c$, $\mathcal{K}''$, $\mathcal{D}''_\pm$, and $Z''(t)$. In this document those double primes are suppressed: every occurrence of $\mathcal{K}$ or $Z(t)$ refers to the final reduced positive-energy interaction space and semigroup, not to the raw indefinite space.

5. GENERATOR AND RESOLVENT

Definition 5.1 (Ordinary semigroup generator). The generator of Z is the closed densely defined operator

$$Bf := \lim_{h \downarrow 0} \frac{Z(h)f - f}{h},$$

with domain

$$\text{Dom } B := \left\{ f \in \mathcal{K} : \lim_{h \downarrow 0} \frac{Z(h)f - f}{h} \text{ exists in } \|\cdot\|_E \right\}.$$

Thus $Z(t) = e^{tB}$ in the C_0 -semigroup sense.

For $\lambda \in \rho(B)$, write

$$R(\lambda, B) := (\lambda I - B)^{-1} \in \mathcal{B}(\mathcal{K}).$$

Because Z is contractive, the Hille–Yosida theorem guarantees $\{\operatorname{Re} \lambda > 0\} \subset \rho(B)$; in particular $R(1, B)$ always exists. The historical criterion instead uses $(B + iI)^{-1} = -R(-i, B)$, whose minus sign is irrelevant to the norm.

Hypothesis 5.2 (Historical resolvent point). For the convention-fixed physical generator B , one has $-i \in \rho(B)$.

Hypothesis 5.2 follows from the full model/resolvent theorem if that theorem gives the spectrum stated in Hypothesis 8.2. Until that identification is verified, $R(1, B)$ provides a guaranteed convention-free regularizer; Lemma 6.4 shows that the exponent is the same whenever both resolvents exist.

Remark 5.3 (Alternative generator convention). If another source writes $Z(t) = e^{-itG}$, then the generator used here is $B = -iG$. One must transform the resolvent factor at the same time. Statements about $B + iI$ cannot be copied between these conventions without this conversion.

6. THE REGULARIZED NORM

Definition 6.1 (Energy operator norm). For $T \in \mathcal{B}(\mathcal{K})$,

$$\|T\|_{E \rightarrow E} := \sup_{0 \neq f \in \mathcal{K}} \frac{\|Tf\|_E}{\|f\|_E}.$$

This is an operator norm on the reduced interaction Hilbert space. It is not the value of the energy quadratic form and not a fixed-vector norm.

Definition 6.2 (Regularized Lax–Phillips quantity). For $t \geq 0$ define

$$(5) \quad M(t) := \left\| Z(t)(B + iI)^{-1} \right\|_{E \rightarrow E},$$

$$(6) \quad \omega_{\text{reg}}(B) := \limsup_{t \rightarrow \infty} \frac{1}{t} \log M(t).$$

The order of the two factors is immaterial because a C_0 -semigroup commutes with every resolvent of its generator:

$$Z(t)(B + iI)^{-1} = (B + iI)^{-1}Z(t).$$

Proposition 6.3 (Exact graph-norm interpretation). *Define*

$$\|f\|_{B,i} := \|(B + iI)f\|_E, \quad f \in \operatorname{Dom} B.$$

Then $\|\cdot\|_{B,i}$ is equivalent to the usual graph norm $\|f\|_E + \|Bf\|_E$, and

$$(7) \quad M(t) = \|Z(t)\|_{(\operatorname{Dom} B, \|\cdot\|_{B,i}) \rightarrow (\mathcal{K}, \|\cdot\|_E)} = \sup_{0 \neq f \in \operatorname{Dom} B} \frac{\|Z(t)f\|_E}{\|(B + iI)f\|_E}.$$

Proof. The map $B + iI : (\operatorname{Dom} B, \|\cdot\|_{B,i}) \rightarrow \mathcal{K}$ is an isometric bijection. Substituting $h = (B + iI)f$ in (5) proves (7). The upper comparison with the standard graph norm is immediate. Conversely, for $h = (B + iI)f$,

$$f = (B + iI)^{-1}h, \quad Bf = h - if,$$

so $\|f\|_E + \|Bf\|_E \leq (1 + 2\|(B + iI)^{-1}\|)\|h\|_E$. Hence the norms are equivalent. $\square$

Lemma 6.4 (The resolvent point is inessential). *For fixed $\lambda, \mu \in \rho(B)$, the functions*

$$M_\lambda(t) := \|Z(t)R(\lambda, B)\|_{E \rightarrow E}, \quad M_\mu(t) := \|Z(t)R(\mu, B)\|_{E \rightarrow E}$$

differ by multiplicative constants independent of t . Consequently their logarithmic limsup exponents are equal.

Proof. The operator

$$C_{\lambda\mu} := (\lambda I - B)(\mu I - B)^{-1} = I + (\lambda - \mu)R(\mu, B)$$

is boundedly invertible. The identity $R(\mu, B) = R(\lambda, B)C_{\lambda\mu}$ and commutation with $Z(t)$ give

$$M_\mu(t) \leq \|C_{\lambda\mu}\| M_\lambda(t).$$

Interchanging λ and μ gives the reverse comparison. $\square$

Remark 6.5. The regularizer is therefore not the special number i ; it is the insertion of one fixed generator resolvent, equivalently one graph norm of regularity. It leaves the output energy norm unchanged.

7. THE QUOTED RH-EQUIVALENT CRITERION AND ITS LOGICAL STATUS

Definition 7.1 (Regularized asymptotic statement). For the convention-fixed modular automorphic Lax–Phillips semigroup, let PF_{reg} denote the assertion

$$(8) \quad \limsup_{t \rightarrow \infty} \frac{1}{t} \log \|Z(t)(B + iI)^{-1}\|_{E \rightarrow E} = -\frac{1}{4}.$$

Quoted Pavlov–Faddeev equivalence. A 2017 retrospective explicitly attributes to the 1972/1975 Pavlov–Faddeev paper the statement

$$(9) \quad \boxed{\text{RH} \iff \text{PF}_{\text{reg}}}.$$

See [5, §3.2, formula (3.6), p. 1004].

This document records (9) as a *source claim*, not as a theorem rederived from the native definitions and proof in [1]. A publication should verify the original space, generator convention, hypotheses, and proof, and then establish their equivalence with Hypotheses 3.1, 4.1, 5.2, 8.1, and 8.2.

Under the unconditional lower bound proved conditionally on Hypothesis 8.2 in Section 9, PF_{reg} is equivalent to

$$(10) \quad \text{for every } \varepsilon > 0 \text{ there is } C_\varepsilon < \infty \text{ such that } M(t) \leq C_\varepsilon e^{(-1/4+\varepsilon)t} \quad (t \geq 0).$$

The equivalence between the limsup upper bound and the family of ε -estimates in (10) uses local boundedness: $\sup_{0 \leq t \leq T} \|Z(t)\| < \infty$ for every $T < \infty$, and the regularizer is fixed and bounded. A finite initial interval is therefore absorbed into C_ε . No estimate with a uniform constant at the boundary exponent $-1/4$ is asserted.

Problem 7.2 (Two separate handoff tasks).

- (T1) *Historical verification/reproof:* verify the original Pavlov–Faddeev statement and prove that its native objects are the objects defined here. Reproving $\text{RH} \Rightarrow \text{PF}_{\text{reg}}$ audits the quoted theorem but does not decide RH.
- (T2) *Unconditional decision:* prove PF_{reg} without assuming RH, or produce a rigorous failure of it that is tied to an off-critical-line zero. By the quoted equivalence—conditional on verifying that its native objects satisfy Hypotheses 3.1, 4.1, 5.2, 8.1, and 8.2—this task is exactly the decision of RH.

Warning 7.3 (Retain the limsup). The family $Z(t)(B + iI)^{-1}$ is not itself a semigroup. There is therefore no submultiplicativity argument that automatically turns the limsup into a limit or an infimum over $t > 0$.

8. CONDITIONAL HARDY-SPACE MODEL AND SPECTRAL INPUT

The elementary model below is exact once an intertwining theorem is supplied. That physical-to-model theorem is isolated as Hypothesis 8.2; nothing in this section silently promotes a scattering resonance to a Hilbert-space eigenvalue.

Let

$$\mathbb{C}_+ := \{s \in \mathbb{C} : \operatorname{Re} s > 0\}, \quad H^2(\mathbb{C}_+) := \left\{ f \text{ holomorphic} : \sup_{x>0} \frac{1}{2\pi} \int_{\mathbb{R}} |f(x+iy)|^2 dy < \infty \right\}.$$

Define the candidate reduced zeta model function

$$\Theta_0(s) := \frac{\xi(2s)}{\xi(-2s)} = \frac{\xi(2s)}{\xi(1+2s)}, \quad \operatorname{Re} s > 0,$$

and isolate the function-theoretic input needed to call it inner.

Hypothesis 8.1 (Reduced inner function). The function Θ_0 belongs to $H^\infty(\mathbb{C}_+)$, has unimodular boundary values almost everywhere on $i\mathbb{R}$, has zero mean type and no singular inner factor, and its zeros in $\mathbb{C}_+$, with multiplicity, are exactly $a = \rho/2$ over the nontrivial zeros of ξ .

Under Hypothesis 8.1, set

$$K_0 := H^2(\mathbb{C}_+) \ominus \Theta_0 H^2(\mathbb{C}_+), \quad A_{0,t} := P_{K_0} M_{e^{-ts}}|_{K_0}.$$

Let B_0 generate $A_{0,t}^*$. A zero a of Θ_0 gives a reproducing-kernel eigenvector

$$A_{0,t}^* k_a = e^{-t\bar{a}} k_a, \quad B_0 k_a = -\bar{a} k_a.$$

For $a = \rho/2$, this is

$$\lambda_\rho = -\frac{\bar{\rho}}{2}.$$

The standard modular Eisenstein scattering coefficient is

$$c(u) = \sqrt{\pi} \frac{\Gamma(u - \frac{1}{2}) \zeta(2u - 1)}{\Gamma(u) \zeta(2u)}.$$

In the centered variable $u = 1/2 + s$, direct algebra gives

$$(11) \quad c\left(\frac{1}{2} + s\right) = \frac{s + \frac{1}{2}}{s - \frac{1}{2}} \Theta_0(s).$$

Thus the physical meromorphic coefficient has an exceptional pole at $s = 1/2$, whereas Θ_0 is the pole-cancelled inner factor. The half-plane Blaschke factor

$$b_{1/2}(s) := \frac{s - \frac{1}{2}}{s + \frac{1}{2}}$$

cancels that pole, and (11) says exactly $b_{1/2}(s)c(1/2 + s) = \Theta_0(s)$. A causal characteristic function must be derived after specifying whether it models c , c^{-1} , or a pole-cancelled version. In particular, the reciprocal of $b_{1/2}$ is meromorphic and is not itself an inner factor on $\mathbb{C}_+$.

Hypothesis 8.2 (Physical-to-model and resonance theorem). The following two assertions hold.

(M1) There are a finite-dimensional exceptional space $\mathcal{K}_{\text{exc}}$, a semigroup $Z_{\text{exc}}(t)$ on it, and a boundedly invertible map

$$W : \mathcal{K} \longrightarrow K_0 \oplus \mathcal{K}_{\text{exc}}$$

such that

$$(12) \quad WZ(t)W^{-1} = A_{0,t}^* \oplus Z_{\text{exc}}(t).$$

The exceptional generator has spectral bound at most $-1/2$.

(M2) The generator B_0 has meromorphic resolvent; every nontrivial zeta zero ρ , with its algebraic multiplicity, gives the actual root space at $-\bar{\rho}/2$; and

$$(13) \quad \sigma(B_0) = \left\{ -\frac{\bar{\rho}}{2} : \xi(\rho) = 0 \right\}.$$

There is no additional boundary spectrum.

The final sentence of Hypothesis 8.2 requires proof in any model derivation. For the reduced inner function it is supported by meromorphic continuation across every finite point of $i\mathbb{R}$, absence of finite zero accumulation, absence of a boundary singular inner factor, and zero mean type; one must still cite the relevant model-generator spectral theorem.

An alternative historical parametrization is $(\rho - 1)/2$. The two multisets agree after applying $\rho \mapsto 1 - \bar{\rho}$ to the zeta zeros. Under Hypothesis 8.2, both conventions give

$$(14) \quad s(B) := \sup_{\lambda \in \sigma(B)} \operatorname{Re} \lambda = -\frac{1}{2} \inf_{\xi(\rho)=0} \operatorname{Re} \rho.$$

Warning 8.3 (Effect of the intertwiner). A bounded similarity preserves the exponential exponent because its condition number contributes only $O(1/t)$ after taking logarithms. Exact equality of finite-time norms requires unitarity. The finite exceptional summand cannot change the target exponent because its spectral/growth exponent is at most $-1/2$, strictly to the left of $-1/4$; this finite-dimensional claim must be checked in the chosen full model.

9. WHY THE CRITERION IMPLIES RH

Proposition 9.1 (Eigenvalue lower bound). *Suppose $Bv = \lambda v$ for some nonzero $v \in \operatorname{Dom} B$ and $\lambda \neq -i$. Then*

$$M(t) \geq \frac{e^{t \operatorname{Re} \lambda}}{|\lambda + i|}, \quad \omega_{\operatorname{reg}}(B) \geq \operatorname{Re} \lambda.$$

Proof. Since $(B + iI)^{-1}v = (\lambda + i)^{-1}v$ and $Z(t)v = e^{t\lambda}v$,

$$\frac{\|Z(t)(B + iI)^{-1}v\|_E}{\|v\|_E} = \frac{e^{t \operatorname{Re} \lambda}}{|\lambda + i|}.$$

Take the operator supremum and then the logarithmic limsup. □

Under Hypothesis 8.2, applying Proposition 9.1 to $\lambda_\rho = -\bar{\rho}/2$ gives

$$(15) \quad \omega_{\operatorname{reg}}(B) \geq -\frac{1}{2} \operatorname{Re} \rho \quad \text{for every nontrivial zero } \rho.$$

The zero set is nonempty and symmetric under $\rho \mapsto 1 - \bar{\rho}$. For each symmetric pair, at least one member has real part at most $1/2$, so (15) gives $\omega_{\operatorname{reg}}(B) \geq -1/4$ unconditionally. If $\omega_{\operatorname{reg}}(B) \leq -1/4$, then (15) excludes every zero with $\operatorname{Re} \rho < 1/2$. The symmetry $\rho \mapsto 1 - \bar{\rho}$ then excludes every zero with $\operatorname{Re} \rho > 1/2$. Hence all nontrivial zeros lie on the critical line.

Thus the implication

$$\omega_{\operatorname{reg}}(B) \leq -\frac{1}{4} \implies \text{RH}$$

uses only the actual-eigenvector part of Hypothesis 8.2 and is not the difficult analytic direction. A pole of a meromorphically continued scattering coefficient would not by itself suffice; the conversion to an eigenvector in $\operatorname{Dom} B$ is essential.

10. WHAT AN UNCONDITIONAL SOLUTION MUST ESTABLISH

Problem 10.1 (Primary solver handoff). First verify Hypotheses 3.1, 4.1, 5.2, 8.1, and 8.2 for the native Pavlov–Faddeev semigroup. Then, without assuming RH, prove

$$(16) \quad \forall \varepsilon > 0 \exists C_\varepsilon < \infty \forall t \geq 0 : \quad \left\| Z(t)(B + iI)^{-1} \right\|_{E \rightarrow E} \leq C_\varepsilon e^{(-1/4 + \varepsilon)t}.$$

Together with the lower bound in Section 9, this proves PF_{reg} , and Proposition 9.1 then proves RH under Hypothesis 8.2.

Conversely, locating a zeta zero off the critical line and applying the functional-equation symmetry produces a zero with real part below $1/2$; that symmetric zero immediately violates (16) through (15) and disproves RH. Merely showing that the operator norm in (16) is too large for a non-spectral, pseudospectral reason would instead refute the quoted implication $\text{RH} \Rightarrow \text{PF}_{\text{reg}}$; it would not, without the verified Pavlov–Faddeev equivalence, disprove RH. This distinction must be maintained in any claimed negative solution.

If the goal is to audit the historical theorem rather than decide RH, the conditional statement to reprove is

$$(17) \quad \text{RH} \implies (16).$$

That implication is nontrivial because the location of $\sigma(B)$ alone does not control a nonnormal semigroup uniformly.

Plausible analytic frameworks for (16) include:

- (R1) a weighted L^2 resolvent estimate combined with Plancherel;
- (R2) a Hardy-space or Carleson-embedding estimate for the regularized model multiplier;
- (R3) a dyadic decomposition in spectral height with oscillatory cancellation; or
- (R4) a quantitative root-space synthesis estimate strong enough to control the graph-norm unit ball.

Each route must exploit the explicit automorphic scattering/resolvent formula; compactness, meromorphic continuation, or spectral location alone is insufficient.

Warning 10.2 (Heuristic obstruction to an absolute contour bound). Do not adopt without proof the absolute operator-norm integrability of

$$\frac{d}{d\tau} (R(a + i\tau, B)R(\lambda_0, B)) = -iR(a + i\tau, B)^2 R(\lambda_0, B)$$

on a vertical line $a = -1/4 + \varepsilon$. On a critical-line eigenvector with eigenvalue $\lambda_n = -1/4 + i\eta_n$, one obtains the lower bound

$$\left\| R(a + i\tau, B)^2 R(\lambda_0, B) \right\| \geq \frac{1}{(\varepsilon^2 + (\tau - \eta_n)^2) |\lambda_0 - \lambda_n|}.$$

Together with sufficiently strong distinct-ordinate density and separation estimates, these lower bounds would make the absolute $L^1(\mathbb{R}; \mathcal{B}(\mathcal{K}))$ integral diverge over dyadic blocks. This document does not supply that arithmetic covering lemma, so divergence is recorded as a serious obstruction rather than a proved theorem. A more plausible contour argument retains oscillation or uses an L^2 /admissibility mechanism instead of taking the absolute operator norm before integration.

11. STATEMENTS THAT MUST NOT BE SUBSTITUTED FOR THE TARGET

11.1. **The unregularized operator norm.** Define

$$\|Z(t)\|_{E \rightarrow E} := \sup_{0 \neq f \in \mathcal{K}} \frac{\|Z(t)f\|_E}{\|f\|_E}.$$

The assertion

$$(18) \quad \limsup_{t \rightarrow \infty} \frac{1}{t} \log \|Z(t)\|_{E \rightarrow E} \leq -\frac{1}{4}$$

implies the regularized estimate, up to the fixed factor $\|(B + iI)^{-1}\|$, and is generally a strictly stronger demand. It is not the target in Definition 7.1. A separate theorem for the reduced scalar compressed-translation model gives $\|A_{0,t}\| = 1$ for all $t > 0$, so its unregularized exponent is 0. Exact finite-time norm one transfers to the physical semigroup only through a unitary version of Hypothesis 8.2; a bounded similarity transfers only the exponent. This model result does not conflict with the regularized criterion, because the vectors nearly attaining the unregularized norm may depend on t and escape to high frequency; the resolvent suppresses precisely such graph-norm-unbounded data.

The available 2002 edition of Lax's text displays the unregularized statement as equation (54), calls it sufficient for RH, and attributes necessity to Pavlov–Faddeev [4]. That later attribution is historically in tension with the explicitly resolvent-regularized Pavlov–Faddeev criterion quoted in [5]. Until the original paper is checked, the two should be reported as distinct source statements rather than harmonized by assumption.

11.2. A dense fixed-vector condition. The Lax–Phillips 1980 exposition records the different sufficient condition

$$(19) \quad \limsup_{t \rightarrow \infty} \frac{1}{t} \log \|Z(t)f\|_E < -\frac{1}{4}, \quad f \in \mathcal{D},$$

where $\mathcal{D}$ is a dense no-standing-waves class [3, §6, formula (6.19)]. Its quantifier order is

$$\exists \mathcal{D} \subset \mathcal{K} \text{ dense} \quad \forall f \in \mathcal{D} \quad \limsup_{t \rightarrow \infty} \frac{1}{t} \log \|Z(t)f\|_E < -\frac{1}{4}.$$

There is no supremum over f after t is chosen. Consequently this condition does not imply an operator-norm estimate without additional uniformity and must not be substituted for (8). The strict inequality is part of this separate sufficient condition; it is not the equality/upper-bound form of the regularized operator criterion.

Under Hypothesis 8.2, the strict dense-vector condition cannot in fact hold on a dense set. A known critical-line zero supplies an isolated eigenvalue λ with $\operatorname{Re} \lambda = -1/4$ and nonzero finite-rank Riesz projection P_λ . If a vector has exponential rate strictly below $-1/4$, commutation of P_λ with $Z(t)$ forces $P_\lambda f = 0$. Hence every such vector lies in the proper closed subspace $\ker P_\lambda$, which cannot contain a dense class. Thus the strict historical display is a sufficient how-not-to-prove-RH condition, not an RH-equivalent target in the model used here.

12. LOGICAL DEPENDENCIES AND PROHIBITED SHORTCUTS

An attempted solution should identify which of the following inputs it uses.

- (L1) **Energy construction:** positivity of E on the reduced no-standing-waves space and unitarity of $U(t)$ there.
- (L2) **Scattering construction:** the Lax–Phillips axioms for $\mathcal{D}_\pm$ and the contraction-semigroup property of $Z(t)$.
- (L3) **Model theorem:** a precise unitary or boundedly invertible intertwiner between the physical semigroup and the reduced K_0 realization plus its exceptional finite-dimensional summand, including the adjoint convention.
- (L4) **Resonance theorem:** meromorphic continuation of the generator resolvent and the zeta-zero/eigenvalue correspondence with algebraic multiplicities.
- (L5) **Uniform analytic estimate:** a resolvent, contour, Hardy-space, or admissibility estimate strong enough to prove (16).

The following shortcuts are invalid unless supplemented by new quantitative arguments:

- (S1) $s(B) = -1/4$ does not imply the regularized or unregularized growth estimate for a nonnormal semigroup.
- (S2) Completeness of eigenvectors or generalized eigenvectors does not imply a uniform synthesis bound; a Riesz-basis or comparable quantitative statement would be needed.
- (S3) Compactness or meromorphicity of the resolvent does not control its norm on high vertical lines.
- (S4) A proof for every fixed f does not permit exchanging $\sup_f$ with $\limsup_{t \rightarrow \infty}$.
- (S5) An approximate eigenvector used to prove $\|Z(t)\| = 1$ may depend on t ; it is not a counterexample to the graph-regularized estimate.
- (S6) A Gearhart–Prüss argument for the unregularized semigroup addresses a different norm and cannot simply discard the fixed right resolvent.
- (S7) Multiple zeta zeros produce root spaces and possible polynomial factors in t ; multiplicities must be retained throughout contour and basis arguments.
- (S8) Replacing Z by Z^* preserves operator norms and real spectral bounds but not fixed-vector stability statements. The convention must remain fixed through the proof.

13. ACCEPTANCE CRITERIA FOR A CLAIMED SOLUTION

A proof or disproof handed back against this specification should contain all of the following.

- (A1) It states whether it works on the physical energy space or on a model space and supplies the exact intertwining map if it transfers between them.
- (A2) It uses the positive reduced energy norm, not the raw indefinite form.
- (A3) It estimates exactly $Z(t)(B + iI)^{-1}$, or an explicitly proved equivalent fixed-resolvent/graph-norm quantity.
- (A4) It keeps the order of quantifiers $\forall \varepsilon \exists C_\varepsilon \forall t$.
- (A5) It proves operator-norm uniformity over the whole graph-norm unit ball.
- (A6) It treats the full scattering factor or proves that the omitted finite factor cannot change the logarithmic exponent.
- (A7) It tracks the generator/time/adjoint convention and the map from zeta zeros to resonances.
- (A8) It justifies every contour deformation, boundary limit, sum over poles, and interchange of limits or suprema.
- (A9) It handles zero multiplicities and generalized eigenvectors.
- (A10) It does not infer a growth estimate merely from spectral location.

14. PROVENANCE AND SOURCE STATUS

The regularized formula (8) is quoted as formula (3.6) in the 2017 survey on Faddeev’s scientific work, where it is explicitly attributed to the Pavlov–Faddeev paper [5, 1]. The original 1972/1975 paper should be consulted for a publication-level check of its exact native notation, hypotheses, and proof. This document fixes a mathematically unambiguous target conditional on the five labeled identification hypotheses; it does not claim that freshly fixing notation proves those identifications.

The Lax–Phillips monograph develops the automorphic energy and scattering spaces [2]; the 1980 survey records the separate dense-vector condition [3]. Uetake gives an operator model for the automorphic Lax–Phillips generator and scattering matrix [6]; the exact theorem needed to establish Hypothesis 8.2, with the present sign and exceptional-factor conventions, must be cited explicitly in any final proof. General C_0 -semigroup and resolvent facts used above can be found in [7].

REFERENCES

- [1] B. S. Pavlov and L. D. Faddeev, *Scattering theory and automorphic functions*, J. Soviet Math. **3** (1975), no. 4, 522–548; English translation of Zap. Nauchn. Sem. LOMI **27** (1972), 161–193.

- [2] P. D. Lax and R. S. Phillips, *Scattering Theory for Automorphic Functions*, Annals of Mathematics Studies, vol. 87, Princeton University Press, Princeton, NJ, 1976.
- [3] P. D. Lax and R. S. Phillips, *Scattering theory for automorphic functions*, Bull. Amer. Math. Soc. (N.S.) **2** (1980), no. 2, 261–295, doi:[10.1090/S0273-0979-1980-14735-7](https://doi.org/10.1090/S0273-0979-1980-14735-7).
- [4] P. D. Lax, *Functional Analysis*, Wiley-Interscience, New York, 2002; see §37.9, equation (54), p. 511.
- [5] L. A. Takhtajan, A. Yu. Alekseev, I. Ya. Aref'eva, M. A. Semenov-Tian-Shansky, E. K. Sklyanin, F. A. Smirnov, and S. L. Shatashvili, *Scientific heritage of L. D. Faddeev. Survey of papers*, Russian Math. Surveys **72** (2017), no. 6, 977–1081; see §3.2, formula (3.6), p. 1004, doi:[10.1070/RM9799](https://doi.org/10.1070/RM9799).
- [6] Y. Uetake, *The Lax–Phillips infinitesimal generator and the scattering matrix for automorphic functions*, Ann. Polon. Math. **92** (2007), no. 2, 99–122, doi:[10.4064/ap92-2-1](https://doi.org/10.4064/ap92-2-1).
- [7] K.-J. Engel and R. Nagel, *One-Parameter Semigroups for Linear Evolution Equations*, Graduate Texts in Mathematics, vol. 194, Springer-Verlag, New York, 2000.